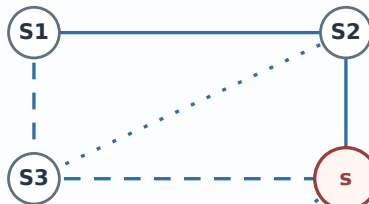
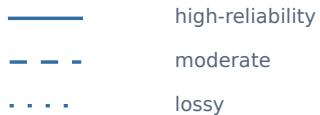
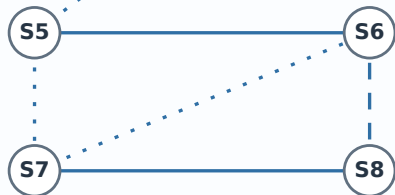


Distributed formation-tracking setting

Formation A



Formation B



Representative node s local processing

s *expanded local view of node s*

Evidence available at node s

Local measurements

measurement update at node s

Local LMB posterior

kinematic states and existence

Neighbor inputs

neighbor LMB posteriors
delivered / dropped counts

Branch-decoupled adaptive fusion

Quality cues

covariance concentration
realized link quality + existence confidence

Shared + branch scores

normalize and interpolate
weak prior: overlap + nominal drop tier

Spatial path

Gaussian KLA fusion

Existence path

Bernoulli KLA fusion
optional FID-FIA

Fused local posterior

extraction and forwarding